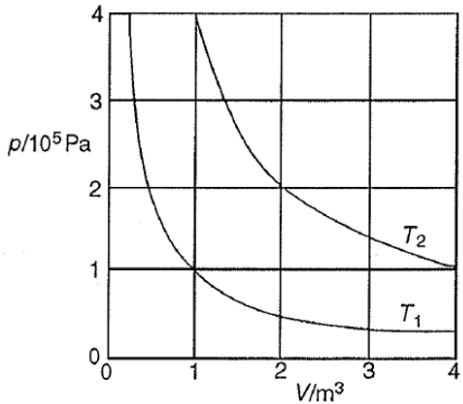
11 The two curves shown below are for a fixed mass of an ideal gas.



What is the ratio $\dfrac{\text{r.m.s. speed of the molecules at temperature } T_2}{\text{r.m.s. speed of the molecules at temperature } T_1}$?

A $\sqrt{2}$ **B** 2 **C** $2\sqrt{2}$ **D** 4